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Team 5

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Use the auto summary option with levels 1, 2 and 3

1 Introduction

1.1 Context

This report converts the flight data sheet into a structured narrative about Azul’s fleet behavior, operational profiles, and route-level efficiency opportunities. The analysis is made in stages, first the raw sample is described, then the typical mission profile of each aircraft is identified, and finally tests are conducted to see if some route assignments could become more fuel efficient without creating a capacity gap.

For this analysis 288,337 flights were used from 1 January 2024 to 15 November 2025. The variables available in the sheet are, flight date and time, origin, destination, equipment type, fuel burn, take-off weight, trip time, and ground distance. Additional metrics were derived to deepen the research, including, route identifiers, trip time in minutes, fuel burned per nautical mile, fuel burned per minute, and fuel burned per ton-nautical mile.

The objective is to understand how the five equipment families in the sample differ in mission length and fuel behaviour, and use those differences to evaluate “sweet spots”, cluster them by behaviour and test for route interchangeability so that fuel could be saved.

2 Data Cleaning

2.1 Exploratory Analysis

The dataset has a lot of variety, the route network covers a total of 419 routes and 113 airports, with 288,337 happening between them. With that data we can compare aircraft behaviour across short-haul and long-haul missions, creating models of every equipment family.

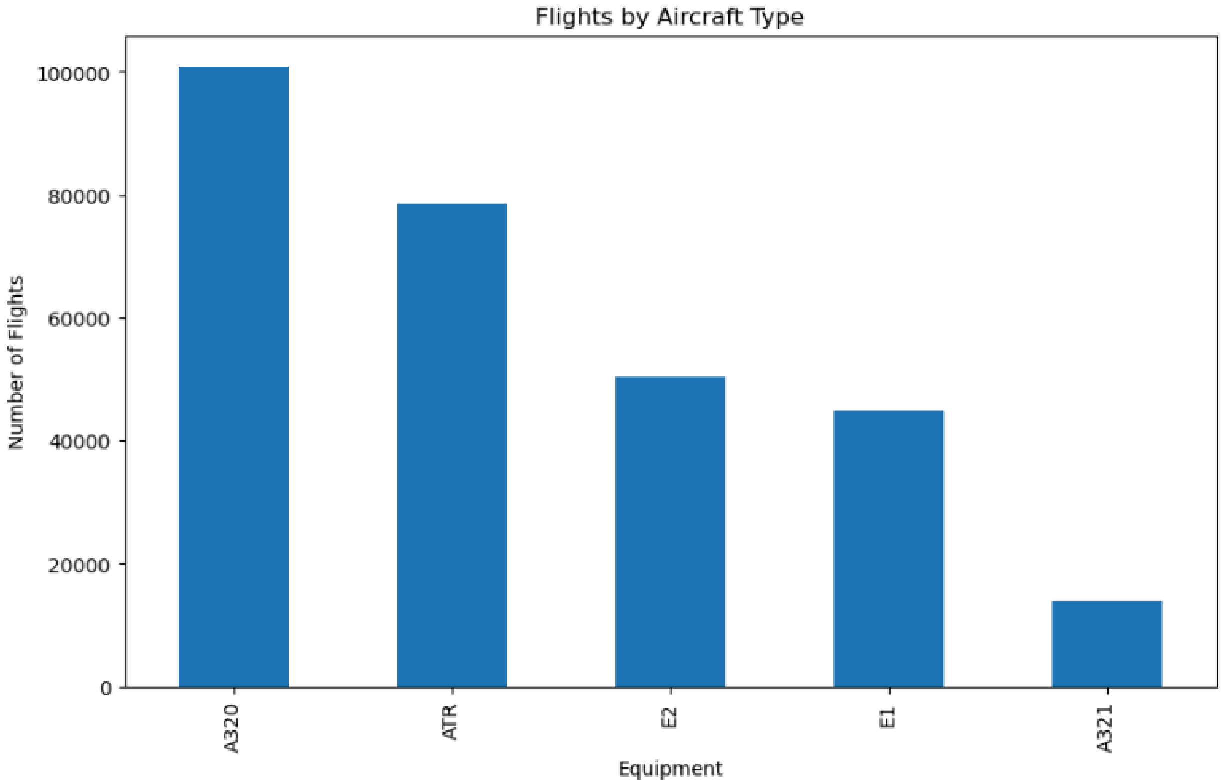

Image 1: Flight count by aircraft type

Although vast, the raw dataset has some problems, analysing closely, unrealistic minimal values for fuel burn and trip time show up in the sheet, including negatives. These values were not ignored in the initial view, as they do make part of the raw sample and help document data conditions. However, for all efficiency-oriented calculations, the values were filtered to keep only positive fuel, trip time, distance and TOW. As they would distort fuel per-distance-indicators.

The fleet mix is concentrated in A320s, which represents 35.0% of flights, ATRs are second with 27.2%. E1 and E2 respectively being 15.5% and 17.4% and A321 being only 4.8% of the sample.

2.2 Operational Profile of the Fleet

Equipment	Flights	Unique routes	Median distance (NM)	Median trip time (min)	Median fuel (kg)
ATR	78,520	220	231.0	62	660

E1	44,816	269	295.6	53	1,994
E2	50,314	260	382.8	64	1,977
A320	100,893	294	889.2	128	4,529
A321	13,794	96	1,170.4	167	7,116.5

Table 1: Fleet operational profile by equipment

The fleet already shows a clear mission ladder. ATR is used on the shortest sectors, with a median distance of 231 NM and median trip time of 62 minutes. E1 and E2 are quite similar and occupy the regional jet space although E2 flies longer sectors on average. A320 and A321 dominate longer stage lengths with median distances of 889.2 NM and 1,170.4 NM respectively.

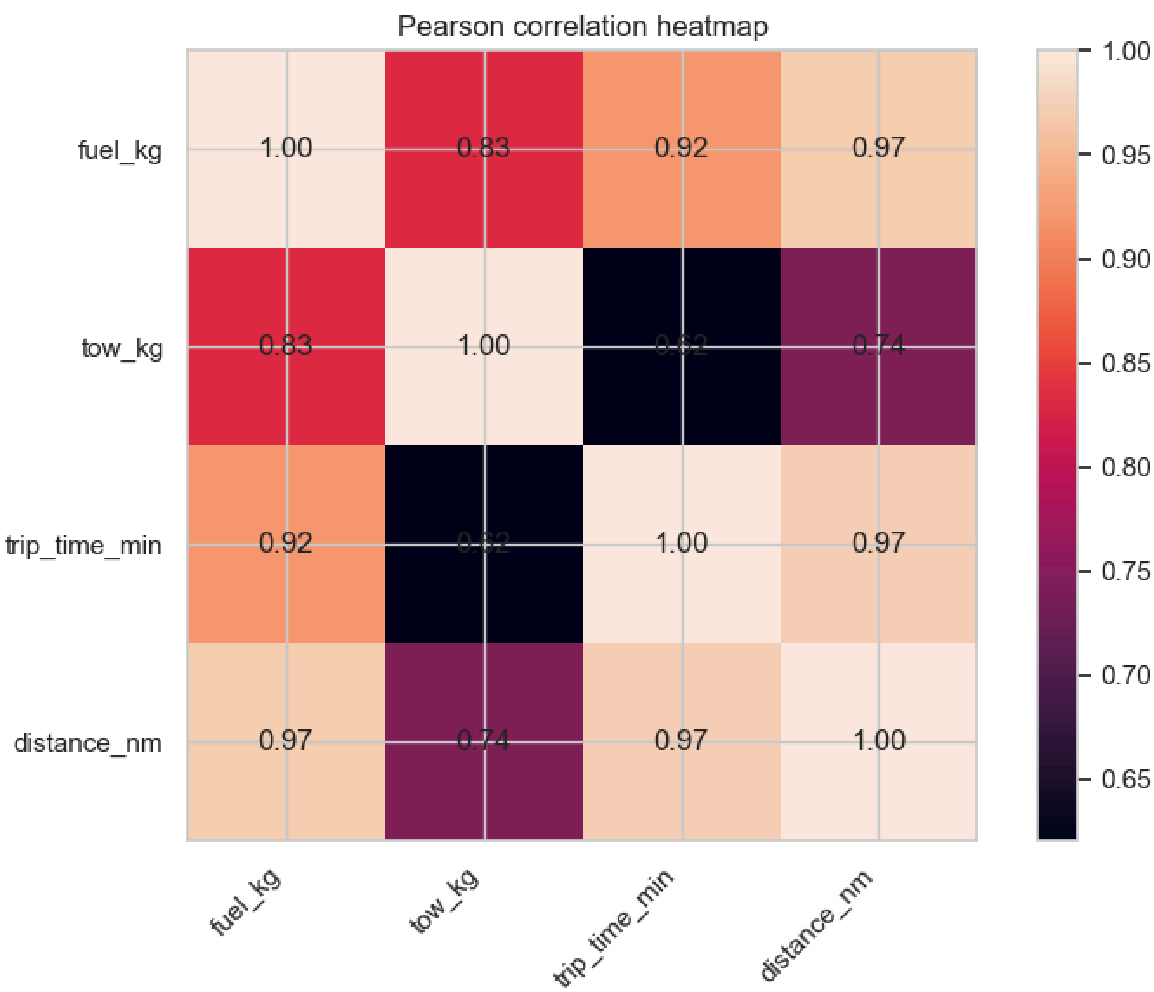

Image 2: Pearson correlation heatmap for the main numerical variables

At aggregate level, distance and trip time are strongly related, fuel burn shows a very strong relationship with both distance and trip time, TOW is still important but it shows less impact than mission length. These patterns instigated the decision to define aircraft behaviour mainly through stage length and fuel efficiency.

2.3 Conclusions

The data-cleaning stage leads to three conclusions. First, the sample is broad enough for comparative analysis. Second, raw outliers and implausible negative operational values exist and must be filtered before any efficiency metric is interpreted. Third, the fleet is already separated into distinct mission layers, which supports a deeper study of aircraft specialization, overlap, and route-level substitution opportunities.

3 Project Scheduling

3.1 Project Planning/Gantt chart

Following the first progress check, we will adjust the presentation to focus on our in-depth investigation of fuel efficiency. Image 3 shows our Gantt chart, which concludes just before the Illinois' students finals. This does not necessarily mean the presentation will be complete, or that Illinois students will stop work. The goal is to finish any analytical work relating to the data before Illinois students finish their semester. After week 6 the work should be primarily preparing the presentation for the in-person presentation to Azul and practicing presentation skills to ensure the presentation runs smoothly.

Image 3: Gantt Chart

3.2 Possible solutions and next steps

To make sure that each goal is met and the overall project is completed to a highly satisfactory standard, the Gantt chart above will be utilized and reviewed as much as possible to hold the team accountable to the set deadlines. As each task is completed, the corresponding cells will be highlighted in order to leave no questions about what has been done and what needs to be done further. This efficiency in our workflow is required to keep the team moving forward at a steady pace.

As the time given in class may not always be sufficient to complete all tasks, our team will need to meet outside of class hours to communicate ideas and complete tasks in a timely and efficient manner. The extracurricular meetings will be hosted either on Zoom or Whatsapp, which are the primary and secondary methods of communication between the American and Brazilian students. Our team will meet at least once a week outside of class time to go over any changes made or to have a general work session. Meeting times will be determined ahead of time based on the availability of all members.

4 Preliminary Data Analysis

4.1 Scope and methods of analysis

Preliminary data analysis used raw data, including outliers to give an understanding of the data set and characterize the nature of the project. The following analysis was done with python scripting and Github for shared access to notebook (explanation of dependencies in section 5). The first steps in understanding the data was to extract the most basic statistics of each flight characteristic given (fuel burn, tow kg, trip time, distance nm), showing mean, standard deviation and quartiles for each characteristic. This led to standard deviations being very large, on account of each aircraft operating in very different conditions, prompting future investigation to be split by aircraft type. Rerunning the previous basic statistical analysis of the flights broken up by aircraft led to much more meaningful results - showing the average performance of each aircraft and allowing us to investigate which aircraft are more efficient in different contexts. During the preliminary analysis, we also began to analyze the fluctuations of in-demand flights throughout the year, with basic graphs of the number of flights month-to-month. The work done with the preliminary analysis was insightful and helped the group to understand the data in a more meaningful way, but can be further analyzed to find useful improvements to Azul's network.

4.2 Results and discussion

The graphs below are some of the most insightful visualizations of the data that we were able to compile. The information from these graphs allowed us to immediately begin brainstorming solutions to Azul's question of fuel consumption. Our python scripts allowed us to organize the data along different useful ranges and domains for analysis. Shown below is the graph for monthly flight volume.

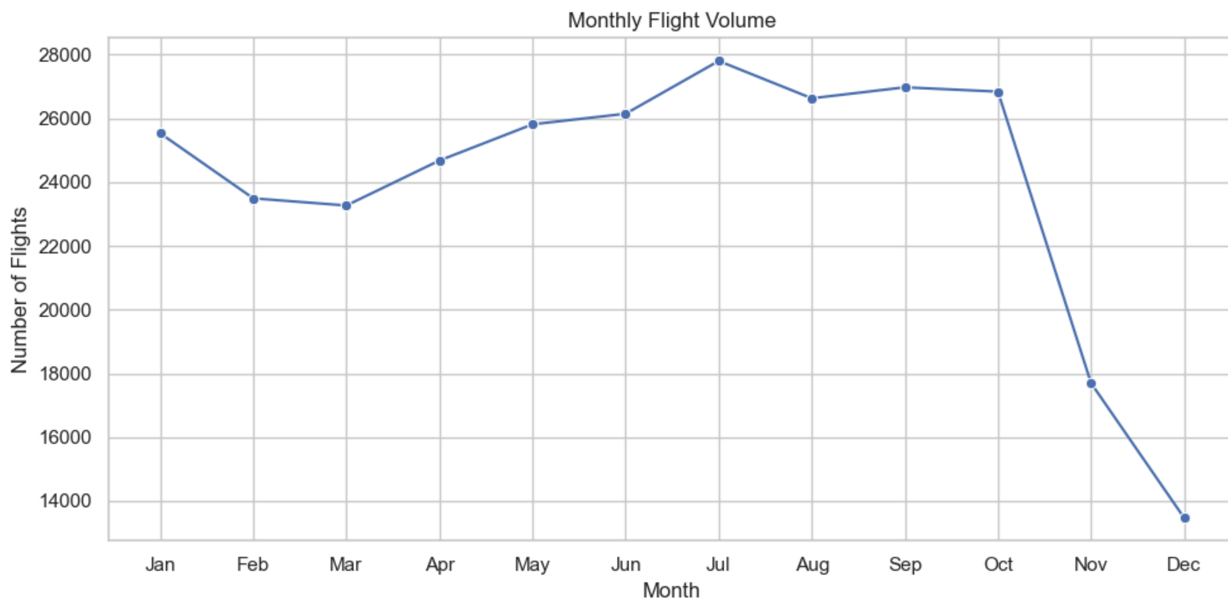

Image 4: Flight Volume by Month

The monthly flight volume graph allows us to observe the trends throughout the year which can help us correlate the yearly events/weather patterns to when people want to travel. Knowing when you will not have as many passengers can allow you to shave off excess flights and condense more people onto larger, more efficient aircraft to cut down on the amounts of flights operating on certain routes.

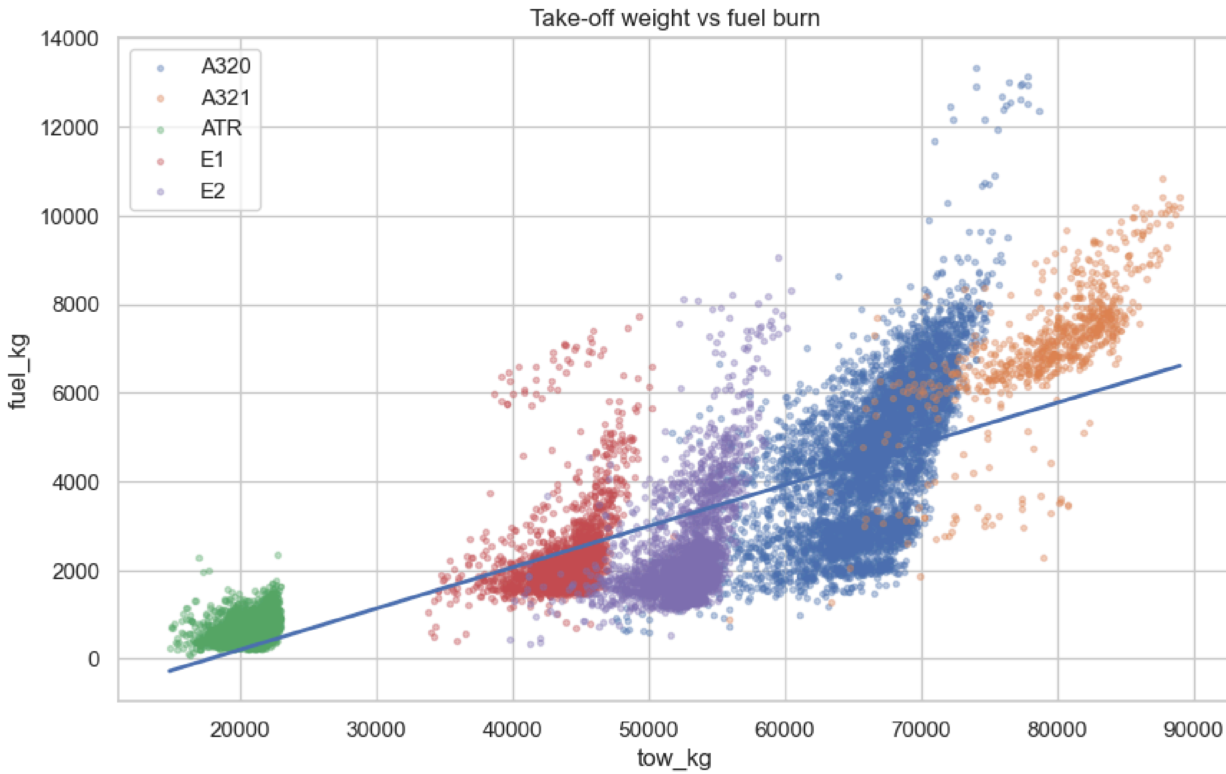

Image 6: Take Off Weight vs Fuel Burn by Aircraft

The key take-aways from images 6 are the groups of outliers that are situated near the average operating conditions of another aircraft. In image 6, there are several flights by the A320 operating at 80 Mg, and burning much more fuel than an A321 would for the same take off weight. This pattern can be seen for several other flights by other aircraft. Identifying the flights that cause these patterns and investigating whether or not the aircraft could be changed for another could lead to reduced fuel consumption.

Our preliminary findings have formed a strong base to understand the dataset and give direction for future research, helping to form strong research questions later. We will expand these analyses to consider which individual routes can actually be made more efficient. The further research will also allow us to create predictive models that will be useful to analyze whether the changes that we implement will produce a significant positive impact on Azul's fuel efficiency.

5 Detail Study

5.1 Proposed research questions

5.2 Methods of analysis

5.3 Results and discussion